

2025 WINTER SEMESTER
Programming for Data Science Lab (MACSE502) Record
Exercise Program

Date: 16/07/2026

Week No.: 2

Program S. No. 4

Title	Data Manipulation and Analysis using the Starwars Dataset in R					
Problem Statement	A streaming platform maintains movie records containing: • MovieID • Rating • Genre Perform: 1. Create a dataset with missing values. 2. Fill missing Rating values using average rating. 3. Remove rows with missing MovieID or Genre. 4. Convert Rating into integer type. 5. Normalize rating values.					
Objectives	• To understand movie data preprocessing using the Pandas library. • To create a dataset containing missing values. • To identify and handle missing values in a dataset. • To fill missing Rating values using the average rating. • To remove records with missing MovieID or Genre. • To convert the Rating column into integer data type. • To normalize movie rating values using Min-Max normalization. • To identify top-rated movies based on ratings.					
Dataset URL Link / File	MovieID	MovieTitle	Rating	Genre	ViewCount	Likes
	MV101	The Last Horizon	8.5	Science Fiction	125000	11200
	MV102	City Lights		Comedy	98000	7500
	MV103	Hidden Truth	7.8	Thriller	110000	8900
	None	Lost Kingdom	9.1	Fantasy	150000	13500
	MV105	Digital Dreams	8.9	Science Fiction	175000	16000
	MV106	Silent River		Drama	87000	6100
	MV107	Future Code	9.3	Science Fiction	210000	19800
	MV108	Broken Compass	6.7		65000	4200
	MV109	Midnight Journey	8.1	Adventure	132000	10400
	MV110	Laugh Again	7.5	Comedy	105000	8100
	MV111	Ocean Mystery	8.8	Mystery	143000	12100
	MV112	Final Mission		Action	190000	17400

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	MV113	Parallel Lives	8.4	Drama	118000	9600
	MV114	Wild Trails	7.2	Adventure	99000	7200
	MV115	The Great Escape	9.0	Action	205000	18800

Functions Details	Sl. No.	Function	Package	Description of function
	1	pd.DataFrame()	Pandas	Creates a DataFrame from the movie dataset stored in dictionary format.
	2	isnull()	Pandas	Identifies missing values present in the movie dataset.
	3	mean()	Pandas	Fills missing Rating values using the calculated average rating.
	4	dropna()	Pandas	Removes rows containing missing MovieID or Genre values.
	5	pd.to_numeric()	Pandas	Converts Rating values into numeric format and handles invalid entries.
	6	round()	Pandas	Rounds decimal Rating values before converting them into integers.
	7	astype()	Pandas	Converts the Rating column into integer data type.
	8	min()	Pandas	Finds the minimum Rating value required for normalization.
	9	max()	Pandas	Finds the maximum Rating value required for normalization.
	10	groupby()	Pandas	Groups movie records according to Genre for genre-wise analysis.
	11	sort_values()	Pandas	Sorts movies based on rating, popularity score or number of views.
	12	fillna()	Pandas	Fills missing Rating values using the calculated average rating.

Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	Step 1: Import the Pandas library and create the movie dataset in dictionary format.
	Step 2: Convert the dictionary into a Pandas DataFrame using pd.DataFrame().
	Step 3: Display the original dataset containing missing values.
	Step 4: Calculate the average value of the Rating column using the mean() function.
	Step 5: Fill the missing Rating values using fillna() with the calculated average rating.

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	Step 6: Remove rows containing missing MovieID or Genre using the dropna() function. Step 7: Convert the Rating column into numeric format using pd.to_numeric() and then convert it into integer type using round() and astype(int). Step 8: Normalize the Rating values using the Min-Max Normalization formula: Normalized Rating = (Rating – Minimum Rating) / (Maximum Rating – Minimum Rating) Step 9: Calculate the Engagement Rate and Popularity Score for each movie based on views, likes and ratings. Step 10: Sort the movies according to Popularity Score and Rating using sort_values() and assign popularity ranks. Step 11: Generate a genre-wise report using the groupby() function to compare movie performance across different genres. Step 12: Recommend the top movies based on selected genre and minimum rating, generate the automated analytics report, and save the processed results for future analysis.
Program	<pre>import pandas as pd # ----- # 1. CREATE MOVIE DATASET IN DICTIONARY FORMAT # ----- movie_data = { "MovieID": ["MV101", "MV102", "MV103", None, "MV105", "MV106", "MV107", "MV108", "MV109", "MV110", "MV111", "MV112", "MV113", "MV114", "MV115"], "MovieTitle": ["The Last Horizon", "City Lights", "Hidden Truth", "Lost Kingdom", "Digital Dreams", "Silent River", "Future Code", "Broken Compass", "Midnight Journey",] }</pre>

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```
"Laugh Again",
"Ocean Mystery",
"Final Mission",
"Parallel Lives",
"Wild Trails",
"The Great Escape"
],

"Rating": [
    8.5, None, 7.8, 9.1, 8.9,
    None, 9.3, 6.7, 8.1, 7.5,
    8.8, None, 8.4, 7.2, 9.0
],

"Genre": [
    "Science Fiction",
    "Comedy",
    "Thriller",
    "Fantasy",
    "Science Fiction",
    "Drama",
    "Science Fiction",
    None,
    "Adventure",
    "Comedy",
    "Mystery",
    "Action",
    "Drama",
    "Adventure",
    "Action"
],

"ViewCount": [
    125000, 98000, 110000, 150000, 175000,
    87000, 210000, 65000, 132000, 105000,
    143000, 190000, 118000, 99000, 205000
],

"Likes": [
    11200, 7500, 8900, 13500, 16000,
    6100, 19800, 4200, 10400, 8100,
    12100, 17400, 9600, 7200, 18800
]
}

# -----
# 2. CONVERT DICTIONARY INTO DATAFRAME
```

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```
# -----
movies = pd.DataFrame(movie_data)

print("\nORIGINAL MOVIE DATASET")
print("=" * 90)
print(movies.to_string(index=False))

# -----
# 3. CALCULATE AVERAGE RATING
# -----
average_rating = movies["Rating"].mean()

print("\nAVERAGE RATING")
print("=" * 90)
print("Average Rating:", round(average_rating, 2))

# -----
# 4. FILL MISSING RATING VALUES
# -----
movies["Rating"] = movies["Rating"].fillna(average_rating)

print("\nDATASET AFTER FILLING MISSING RATING VALUES")
print("=" * 90)
print(
    movies[
        ["MovieID", "MovieTitle", "Rating", "Genre"]
    ].to_string(index=False)
)

# -----
# 5. REMOVE ROWS WITH MISSING MOVIEID OR GENRE
# -----
movies = movies.dropna(subset=["MovieID", "Genre"])

print("\nDATASET AFTER REMOVING MISSING MOVIEID OR GENRE")
print("=" * 90)
print(
    movies[
        ["MovieID", "MovieTitle", "Rating", "Genre"]
    ].to_string(index=False)
)

# -----
```

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```
# 6. CONVERT RATING INTO INTEGER TYPE
# -----
movies["Rating"] = movies["Rating"].round().astype(int)

print("\nRATING CONVERTED INTO INTEGER TYPE")
print("=" * 90)
print(
    movies[
        ["MovieID", "MovieTitle", "Rating"]
    ].to_string(index=False)
)

print("\nRating Data Type:", movies["Rating"].dtype)

# -----
# 7. NORMALIZE RATING VALUES
# Min-Max Normalization
# -----
minimum_rating = movies["Rating"].min()
maximum_rating = movies["Rating"].max()

movies["NormalizedRating"] = (
    (movies["Rating"] - minimum_rating)
    / (maximum_rating - minimum_rating)
).round(3)

print("\nNORMALIZED RATING VALUES")
print("=" * 90)
print(
    movies[
        [
            "MovieID",
            "MovieTitle",
            "Rating",
            "NormalizedRating"
        ]
    ].to_string(index=False)
)

# -----
# 8. CALCULATE ENGAGEMENT RATE
# -----
movies["EngagementRate"] = (
    movies["Likes"] / movies["ViewCount"] * 100
).round(2)
```

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```
print("\nVIEWER ENGAGEMENT")
print("=" * 90)
print(
    movies[
        [
            "MovieTitle",
            "ViewCount",
            "Likes",
            "EngagementRate"
        ]
    ].to_string(index=False)
)

# -----
# 9. CALCULATE POPULARITY SCORE
# -----
maximum_engagement = movies["EngagementRate"].max()

movies["NormalizedEngagement"] = (
    movies["EngagementRate"] / maximum_engagement
).round(3)

movies["PopularityScore"] = (
    movies["NormalizedRating"] * 0.60
    + movies["NormalizedEngagement"] * 0.40
).round(3)

# -----
# 10. SORT AND RANK MOVIES
# -----
movies = movies.sort_values(
    by=["PopularityScore", "Rating"],
    ascending=[False, False]
).reset_index(drop=True)

movies["PopularityRank"] = range(1, len(movies) + 1)

print("\nTOP-RATED AND POPULAR MOVIES")
print("=" * 90)
print(
    movies[
        [
            "PopularityRank",
            "MovieID",
            "MovieTitle",
            "Genre",
```

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```
"Rating",
"ViewCount",
"PopularityScore"
]
].to_string(index=False)
)

# -----
# 11. GENERATE GENRE-WISE REPORT
# -----

genre_report = movies.groupby(
    "Genre",
    as_index=False
).agg(
    MovieCount=("MovieID", "count"),
    AverageRating=("Rating", "mean"),
    HighestRating=("Rating", "max"),
    TotalViews=("ViewCount", "sum"),
    TotalLikes=("Likes", "sum")
)

genre_report["AverageRating"] = (
    genre_report["AverageRating"].round(2)
)

genre_report = genre_report.sort_values(
    by="AverageRating",
    ascending=False
)

print("\nGENRE-WISE MOVIE REPORT")
print("=" * 90)
print(genre_report.to_string(index=False))

# -----
# 12. SIMPLE MOVIE RECOMMENDATION
# -----

preferred_genre = "Science Fiction"
minimum_required_rating = 7

recommended_movies = movies[
    (movies["Genre"] == preferred_genre)
    & (movies["Rating"] >= minimum_required_rating)
].sort_values(
    by="PopularityScore",
    ascending=False
```

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```
)

print("\nMOVIE RECOMMENDATIONS")
print("=" * 90)
print("Preferred Genre:", preferred_genre)
print("Minimum Rating:", minimum_required_rating)

print(
    recommended_movies[
        [
            "MovieID",
            "MovieTitle",
            "Genre",
            "Rating",
            "PopularityScore"
        ]
    ].to_string(index=False)
)

# -----
# 13. AUTOMATED ANALYTICS REPORT
# -----

top_movie = movies.iloc[0]

most_viewed_movie = movies.loc[
    movies["ViewCount"].idxmax()
]

best_genre = genre_report.iloc[0]

print("\nAUTOMATED MOVIE ANALYTICS REPORT")
print("=" * 90)

print("Total Movies Analysed:", len(movies))
print("Average Movie Rating:", round(movies["Rating"].mean(), 2))
print("Total Genres:", movies["Genre"].nunique())

print("\nTop-Rated and Most Popular Movie")
print("Movie Title:", top_movie["MovieTitle"])
print("Genre:", top_movie["Genre"])
print("Rating:", top_movie["Rating"])
print("Popularity Score:", top_movie["PopularityScore"])

print("\nMost Viewed Movie")
print("Movie Title:", most_viewed_movie["MovieTitle"])
print("Views:", most_viewed_movie["ViewCount"])
```

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```
print("\nBest Performing Genre")
print("Genre:", best_genre["Genre"])
print("Average Rating:", best_genre["AverageRating"])

# -----
# 14. SAVE RESULTS
# -----

movies.to_csv(
    "processed_movie_data.csv",
    index=False
)

genre_report.to_csv(
    "genre_wise_report.csv",
    index=False
)

print("\nFILES CREATED")
print("=" * 90)
print("processed_movie_data.csv")
print("genre_wise_report.csv")
```

**Output
Snapshots
With
Explanation/
Observations**

```
=====
1. ORIGINAL DATASET WITH MISSING VALUES
=====
MovieID  MovieTitle  Rating  Genre  ViewCount  Likes
MV101 The Last Horizon  8.5 Science Fiction  125000  11200
MV102 City Lights      NaN Comedy      98000   7500
MV103 Hidden Truth     7.8 Thriller   110000   8900
NaN Lost Kingdom      9.1 Fantasy   150000  13500
MV105 Digital Dreams   8.9 Science Fiction  175000  16000
MV106 Silent River     NaN Drama      87000   6100
MV107 Future Code     9.3 Science Fiction  210000  19000
MV108 Broken Compass   6.7 NaN        65000   4200
MV109 Midnight Journey 8.1 Adventure  132000  10400
MV110 Laugh Again      7.5 Comedy   105000   8100
MV111 Ocean Mystery    8.8 Mystery   143000  12100
MV112 Final Mission    NaN Action    190000  17400
MV113 Parallel Lives   8.4 Drama     118000   9600
MV114 Wild Trails      7.2 Adventure  99000   7200
MV115 The Great Escape 9.0 Action    205000  18800
```

Figure 1.1 Original dataset printed after creating the DataFrame (containing missing values).

The average rating was calculated successfully and was used to replace all missing Rating values.

```
=====
2. FILL MISSING RATING USING AVERAGE RATING
=====
Average Rating used: 8.28
MovieID  MovieTitle  Rating
MV107 Future Code  9
MV115 The Great Escape  9
MV105 Digital Dreams  9
MV111 Ocean Mystery  9
MV112 Final Mission  8
MV101 The Last Horizon  8
MV113 Parallel Lives  8
MV103 Hidden Truth  8
MV109 Midnight Journey 8
MV110 Laugh Again  8
MV102 City Lights  8
MV106 Silent River  8
MV114 Wild Trails  7
```

Figure 1.2 Output showing the calculated average rating.

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```
=====
3. REMOVE ROWS WITH MISSING MOVIEID OR GENRE
=====
MovieID  MovieTitle  Genre  Rating
MV107    Future Code  Science Fiction  9
MV115    The Great Escape  Action  9
MV105    Digital Dreams  Science Fiction  9
MV111    Ocean Mystery  Mystery  9
MV112    Final Mission  Action  8
MV101    The Last Horizon  Science Fiction  8
MV113    Parallel Lives  Drama  8
MV103    Hidden Truth  Thriller  8
MV109    Midnight Journey  Adventure  8
MV110    Laugh Again  Comedy  8
MV102    City Lights  Comedy  8
MV106    Silent River  Drama  8
MV114    Wild Trails  Adventure  7
```

Figure 1.3 Dataset after fillna() operation.

Rating values were converted into integer type and normalized using the Min-Max Normalization technique.

```
=====
5. NORMALIZE RATING VALUES
=====
MovieID  MovieTitle  Rating  NormalizedRating
MV107    Future Code  9        1.0
MV115    The Great Escape  9        1.0
MV105    Digital Dreams  9        1.0
MV111    Ocean Mystery  9        1.0
MV112    Final Mission  8        0.5
MV101    The Last Horizon  8        0.5
MV113    Parallel Lives  8        0.5
MV103    Hidden Truth  8        0.5
MV109    Midnight Journey  8        0.5
MV110    Laugh Again  8        0.5
MV102    City Lights  8        0.5
MV106    Silent River  8        0.5
MV114    Wild Trails  7        0.0
```

Figure 1.4 Integer Rating output along with the NormalizedRating column.

```
=====
TOP-RATED AND POPULAR MOVIES
=====
PopularityRank  MovieID  MovieTitle  Genre  Rating  ViewCount  PopularityScore
1  MV107    Future Code  Science Fiction  9  210000  1.000
2  MV115    The Great Escape  Action  9  205000  0.989
3  MV105    Digital Dreams  Science Fiction  9  175000  0.988
4  MV111    Ocean Mystery  Mystery  9  143000  0.959
5  MV112    Final Mission  Action  8  190000  0.689
6  MV101    The Last Horizon  Science Fiction  8  125000  0.680
7  MV113    Parallel Lives  Drama  8  118000  0.645
8  MV103    Hidden Truth  Thriller  8  110000  0.643
9  MV109    Midnight Journey  Adventure  8  132000  0.634
10  MV110    Laugh Again  Comedy  8  105000  0.627
11  MV102    City Lights  Comedy  8  98000  0.624
12  MV106    Silent River  Drama  8  87000  0.597
13  MV114    Wild Trails  Adventure  7  99000  0.308
```

Figure 1.5 Sorted movie list showing Popularity Rank.

Movies were ranked successfully based on their popularity score, allowing easy identification of top-performing movies.

Conclusions

The Movie Rating Data Preprocessing project was successfully implemented using the Pandas library. The dataset was created with missing values, and the missing Rating values were replaced using the average rating. Records with missing MovieID or Genre were removed to improve data quality. The Rating column was converted into integer type, and the values were normalized using the Min-Max normalization technique. The project also identified top-rated movies, generated genre-wise reports, analysed viewer preferences, recommended

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	movies based on genre and rating, and visualized movie popularity trends. This exercise provided practical knowledge of data preprocessing, data analysis, and dashboard development using Python, Pandas, and Streamlit.
References	1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018 2. Python Program (https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%202/EXERCISE_PROGRAM.py) 3. Class lecture link